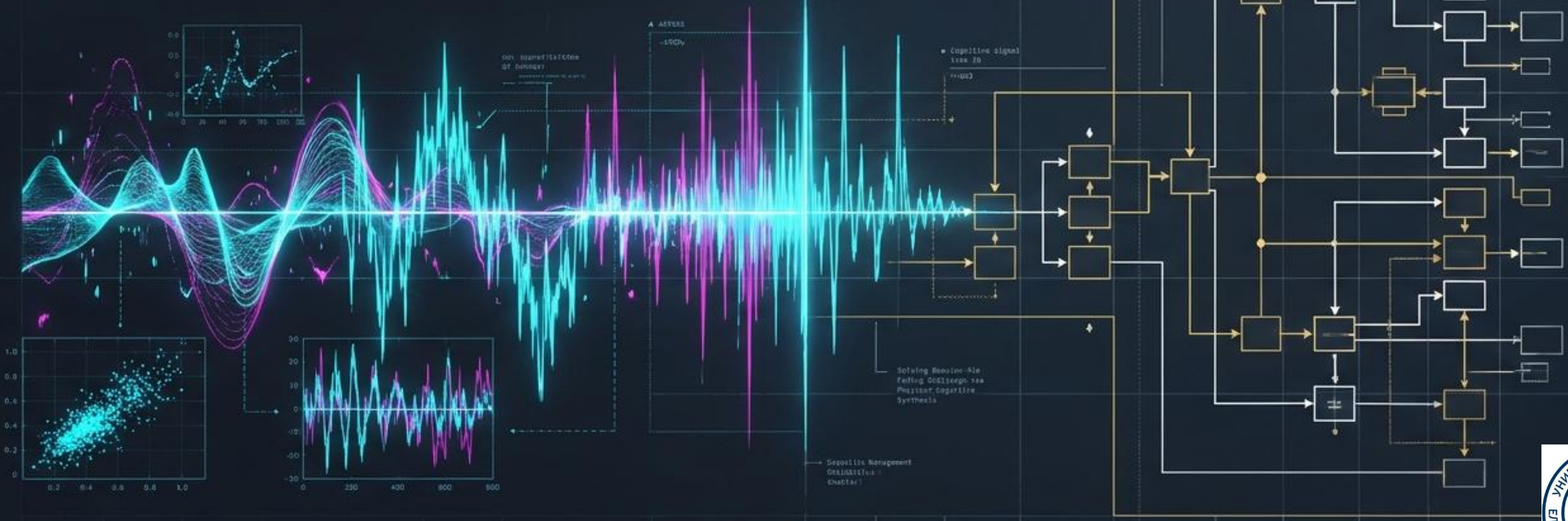


Network Management Reliability Enabled by LLM

Solving the Beaulieu-Xie Fading Challenge
via Physical-Cognitive Synthesis

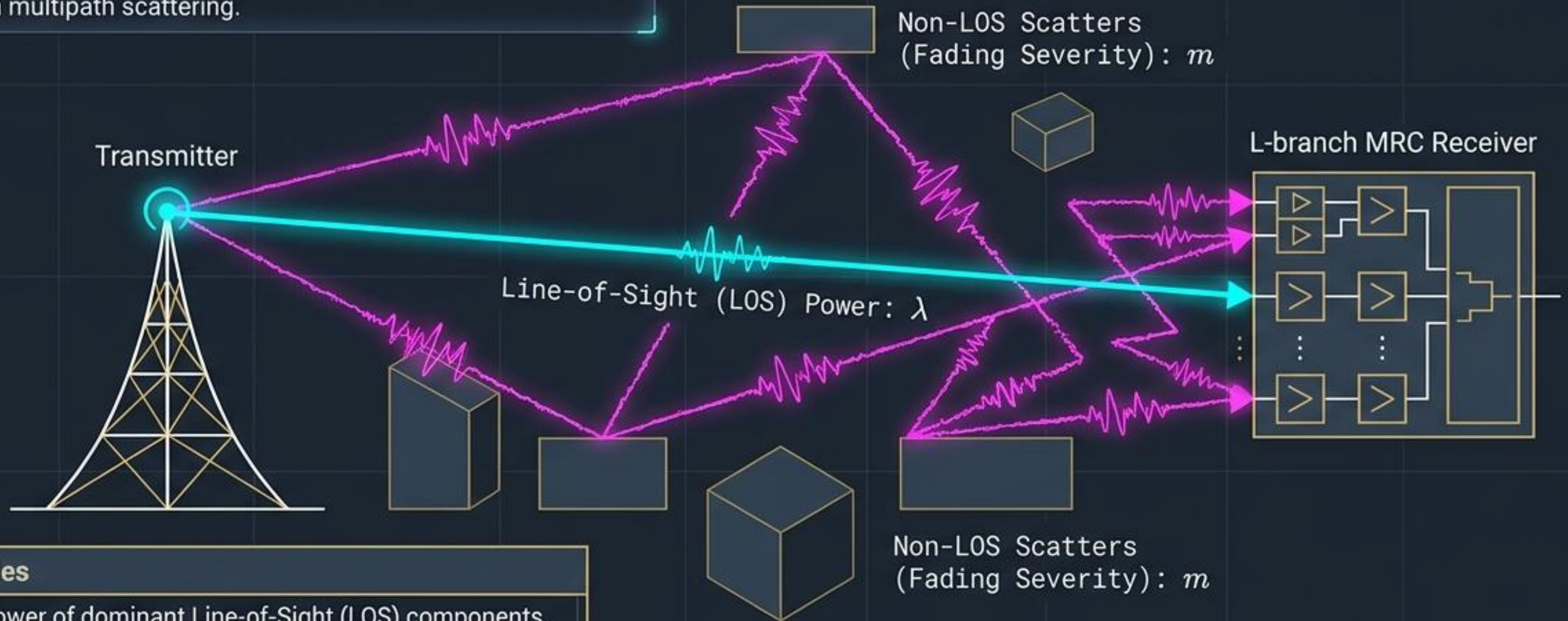


Presented by: Nenad Petrovic and Dragana Krstic



The Physical Context: Beaulieu-Xie Fading

Wireless systems (MIMO, satellite, terrestrial) operate in environments with multiple dominant specular components and random multipath scattering.



Variables

λ :	Power of dominant Line-of-Sight (LOS) components.
m :	Shape parameter quantifying fading severity.
L :	Number of diversity branches in the MRC receiver.

The Problem: The Latency Trap in Network Management

Core Issue: Signal quality degrades rapidly below critical thresholds due to fading, but traditional management mechanisms suffer from delayed response times.

Root Cause: Poor coordination, missing dependencies, and incorrect ordering between performance degradation detection and reliability enhancement procedures.

Result

Result: Reduced achievable throughput and disrupted service continuity.

Physical Reality

Channel Capacity

Threshold

The Latency Trap

Corrective Action Activated

System Response

The Solution: A Two-Tiered Defense Architecture

Tier 2: LLM-Driven Network Management

Gen-AI
logic engine

Tier 2 (Cognition): Optimize the network logically.

A Retrieval-Augmented Generation (RAG) LLM framework automates rule validation and coordinates real-time recovery actions to prevent latency traps.

Tier 1: L-branch MRC Receivers

Real-Time
Capacity
Adjustments

Tier 1 (Physics): Mitigate fading physically.

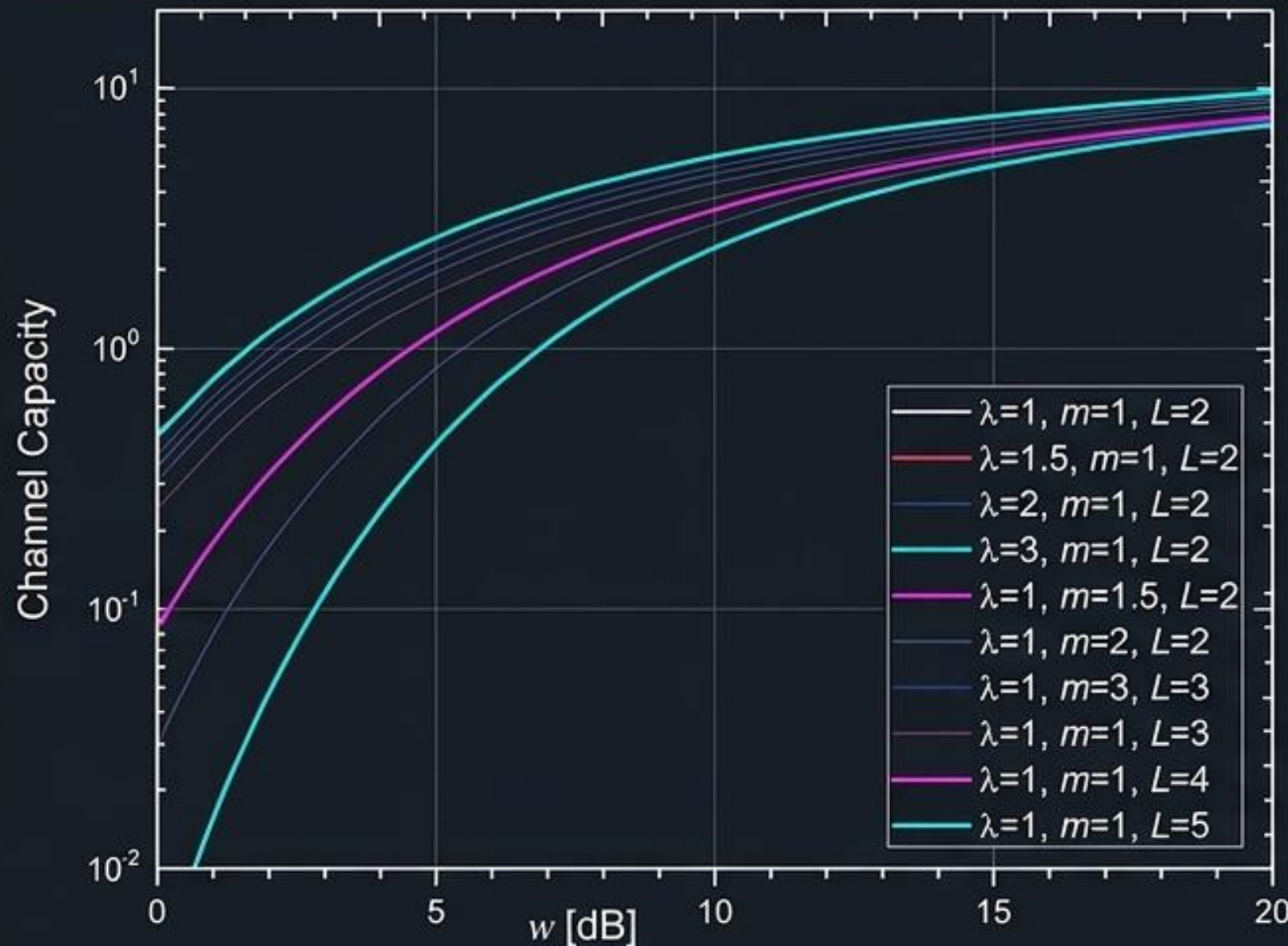
L-branch Maximal Ratio Combining (MRC) receivers combine signals to maximize SNR and improve channel capacity.

Signal-to-Noise Ratio
(SNR) maximization

Channel Capacity
Improvement

Process Stage I: Deriving Physical Capacity

Convergence Chart



1

The Math: Analytically derived Channel Capacity (C/B) under BX fading serves as the primary reliability metric.

$$C/B = 1/\ln(2) \int_0^{+\infty} \ln(1+\gamma) p_{\gamma_i}(\gamma) d\gamma = \frac{1}{\ln(2)} e^{-\frac{\lambda^2 L}{2}} e^{\frac{m_i}{w_i}} \sum_{j_1=0}^{\infty} \sum_{j_2=1}^{j_1+Lm_i} \frac{\lambda^{2j_1}}{j_1!} \cdot \frac{(j_1+Lm_i-1)!}{\Gamma(j_1+Lm_i)} \left(\frac{L}{2}\right)^{j_1} \left(\frac{m_i}{w_i}\right)^{j_1-j_2+Lm_i} \Gamma\left(j_2-j_1-Lm_i, \frac{m_i}{w_i}\right)$$

2

The Physics Insight: Increasing MRC branches (L) drastically mitigates fading severity (m). Higher LOS power (λ) improves propagation but requires more computational terms.

3

The Computational Insight:

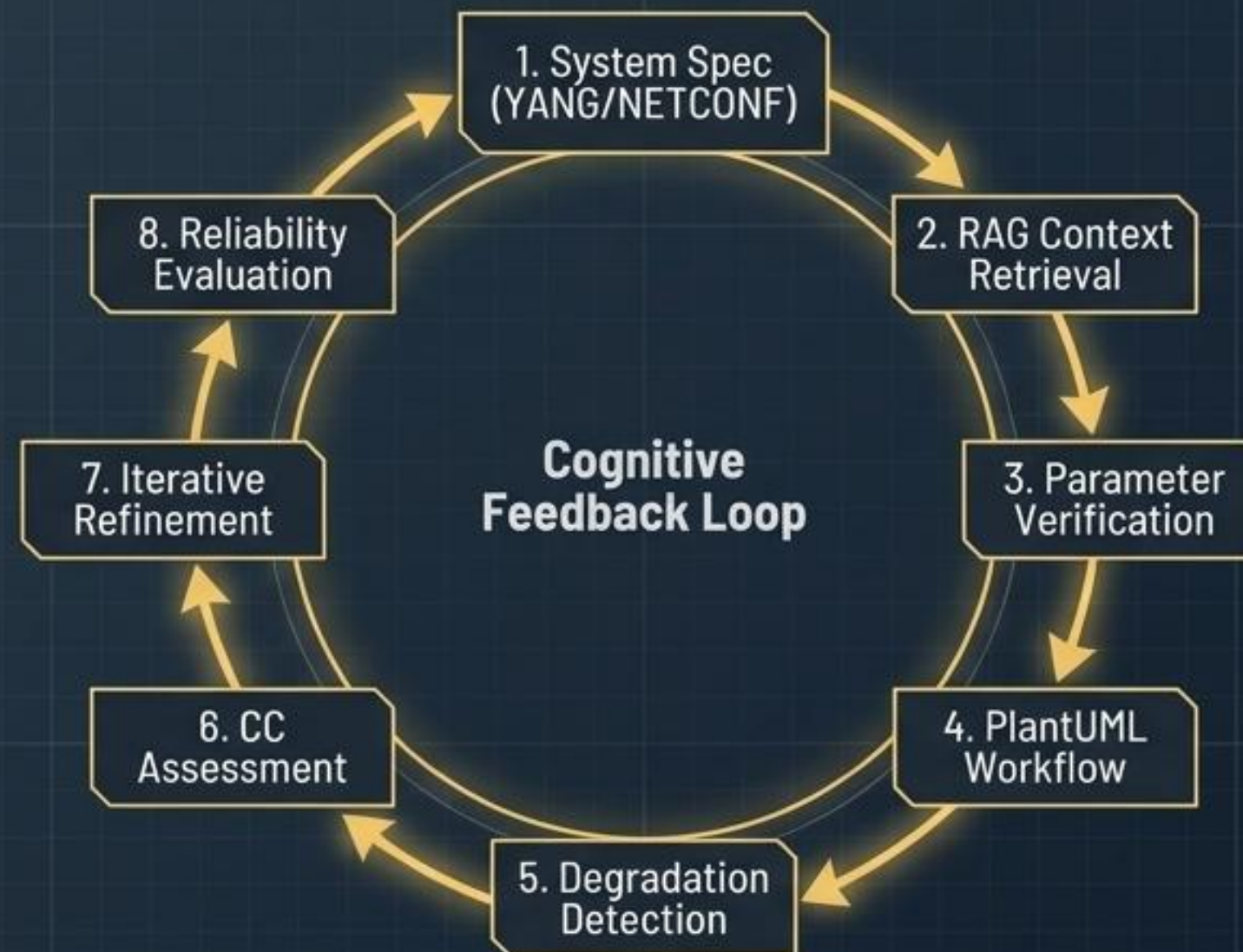
w (dB)	Terms for 5-decimal Accuracy
-10	5

Series expression convergence is highly efficient. At low SNR (w=-10 dB), calculation requires only 5 or terms to achieve five decimal place accuracy.

Process Stage II: The LLM Generation Pipeline

Mechanism: An automated Model-Driven Engineering (MDE) pipeline implemented in Node-RED.

Function: Translates mathematical capacity drops (Stage I) into structured, LLM-verifiable activity workflows (PlantUML) that dictate network behavior.



Human Interaction: The Engineer in the Loop

Human Role

Engineers define management scripts, system models, and role-based reliability rules at design time.

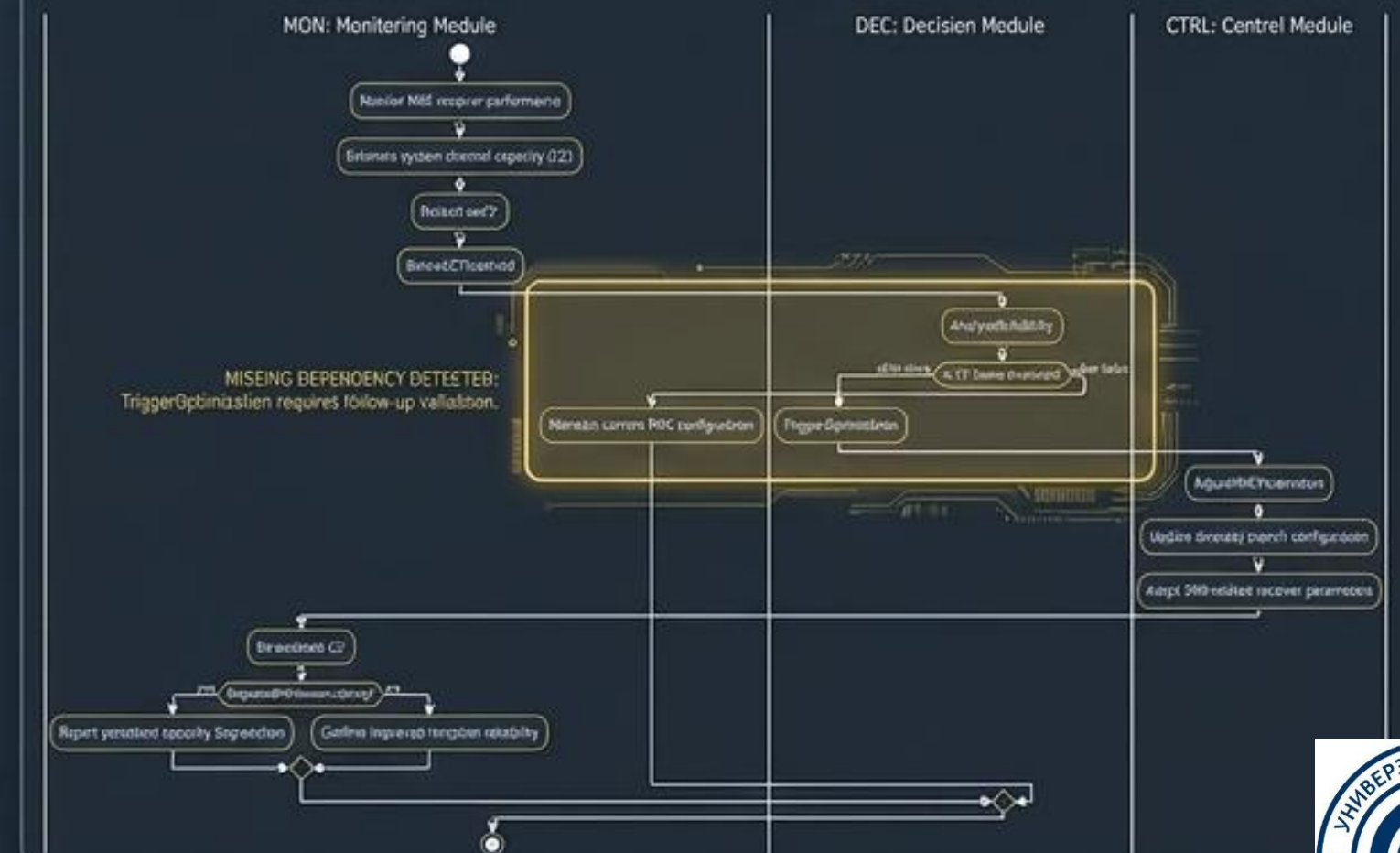
AI Assist

The Gen-AI framework simulates degradation events to automatically detect missing dependencies or incorrect action orderings before physical deployment.

Human Input

```
# System Rule Definitions
RULE_SET communication_reliability {
  CONSTRAINT: Ai precedes/follows Aj;
  CONSTRAINT: (Rk:Am) -> VALIDATE_CONFIG(Am);
  CONSTRAINT: ON_EVENT(degradation) -> TRIGGER_OPTIMIZATION;
  RULE: IF (CC < THRESHOLD) THEN
    TRIGGER(AnalyzeReliability);
  RULE: IF (AnalyzeReliability.RESULT == 'false') THEN
    ADJUST(MRC_Parameters);
  RULE: IF (AnalyzeReliability.RESULT == 'true') THEN
    TRIGGER(Maintain_Configuration);
}
```

AI Output



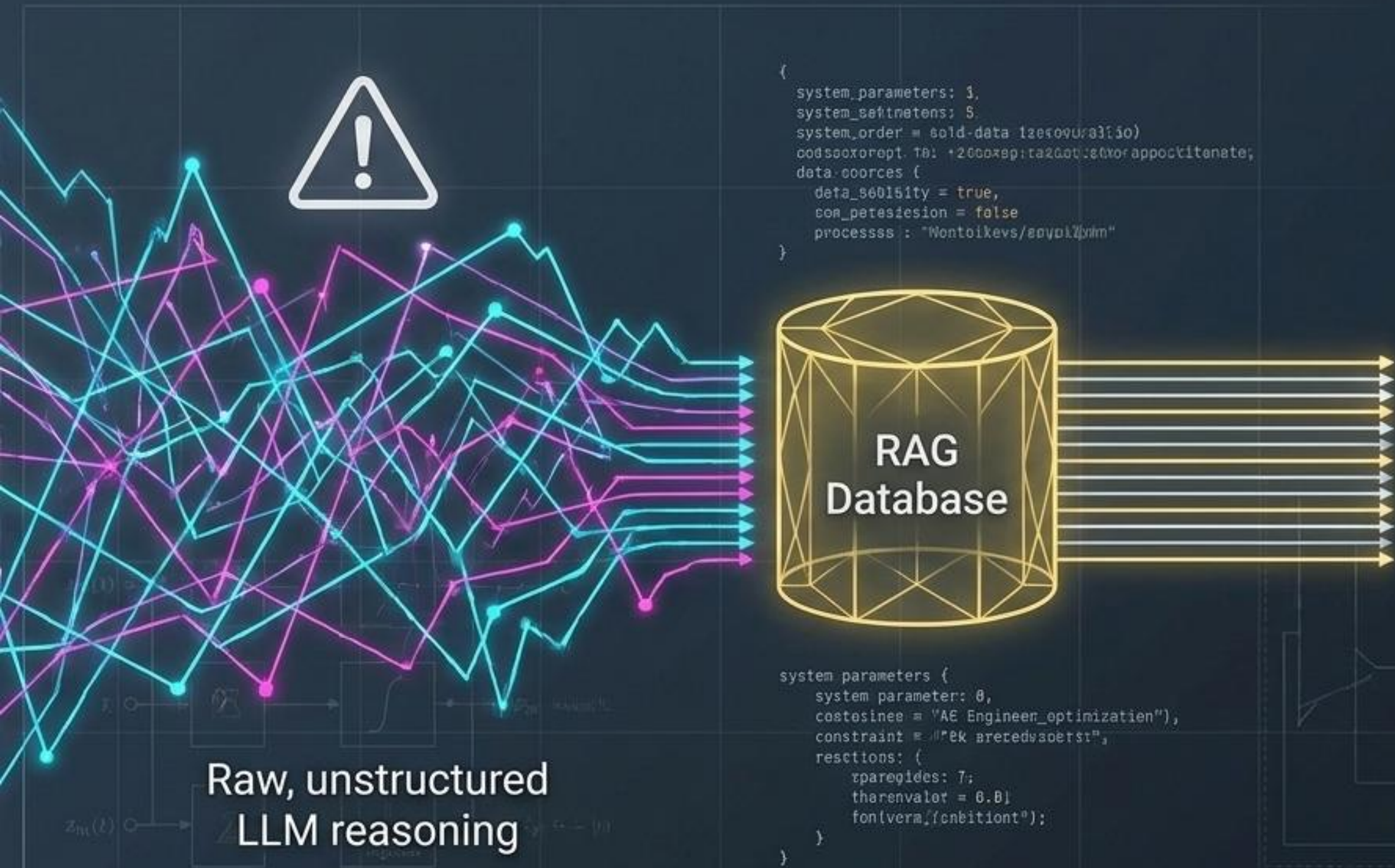
Evaluation: LLM Performance Matrix

Task	GPT-5	GPT-OSS-120B (Local)	Relative Speed-up
Parameter Extraction (MRC, BX fading, CC threshold)	100%	90%	8x
State Modeling (Reliability states, performance drops)	90%	80%	9x
Activity-Driven Reliability Validation	90%	70%	12x

Tested in discrete-event simulation. Inference parameters locked: temperature = 0.1, top-p = 0.9.



Challenges: Context, Coordination, and Complexity



Design Inconsistencies:
Legacy network rules frequently lack properly defined dependencies between monitoring drops and corrective actions.

LLM Limitations:
Open-source models (GPT-OSS-120B) occasionally exhibit parameter redundancy and dependency inconsistencies in complex workflows.

The Mitigation:

Retrieval-Augmented Generation (RAG) is strictly required to constrain LLM hallucinations and maintain precise contextual grounding regarding BX fading mathematics.



The Benefits: A Reliability Multiplier



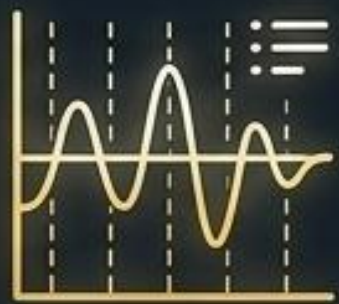
12x Faster

Unprecedented speed-up in design-time, activity-driven reliability validation compared to human experts.



Automated Consistency

Eradicates the latency gap between performance degradation detection and recovery actions.



Maximized Efficiency

Ensures MRC receiver parameters adapt dynamically to preserve Shannon Capacity in severe fading environments.

Transferability: Evolution of the MDE+LLM Framework

IoT Security
(Previous Work)

Origin: Methodology successfully adapted from prior domain research in IoT-based system security.

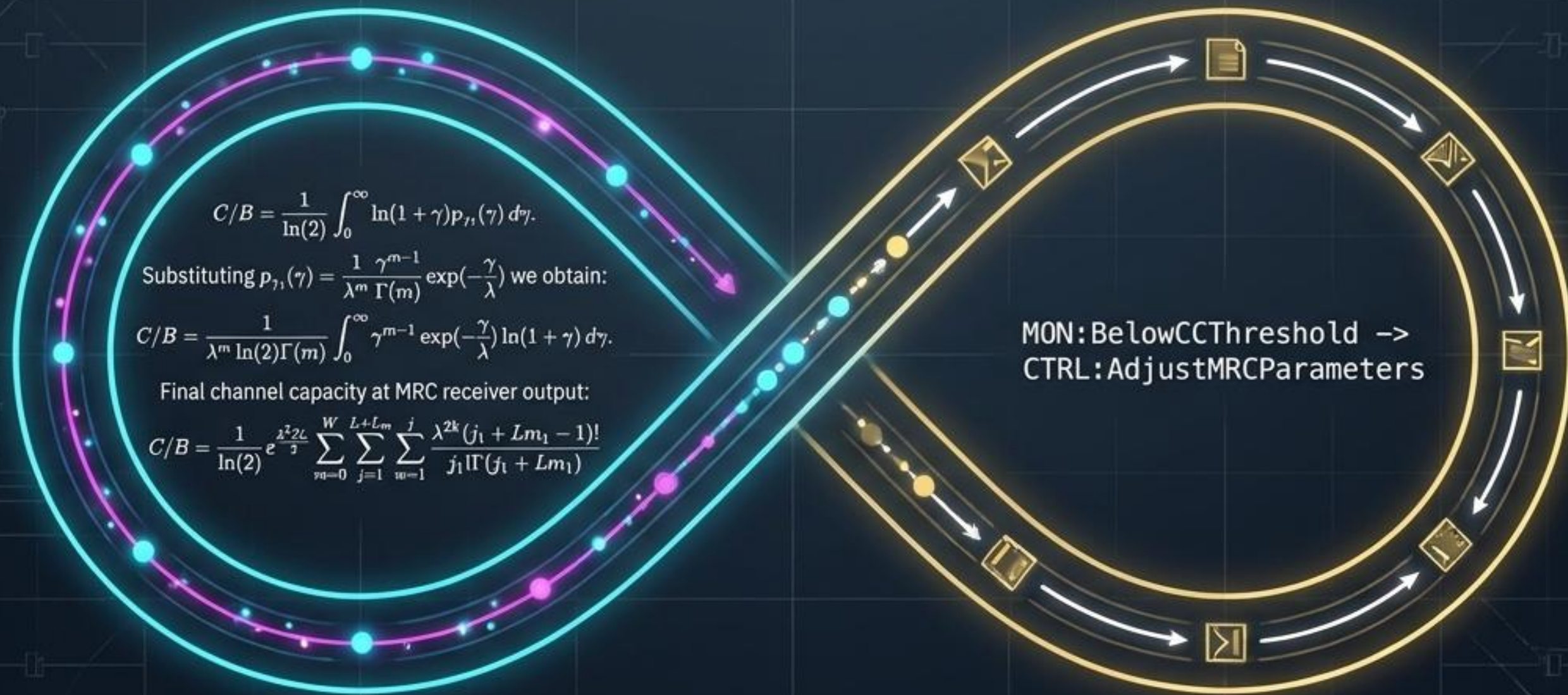
BX Fading Network Capacity
(Current Work)

Current State: Validated for complex, mathematics-heavy physical layer communications.

Agentic GenAI Autonomy
(Future Work)

Future Horizon: Transitioning from design-time validation to run-time Agentic AI, enabling fully autonomous, adaptive network management systems.

Synthesis: The Reliability Loop



Raw physics translated into machine-enforced logic.
By transforming complex fading mathematics into actionable, LLM-verifiable context, we guarantee that network infrastructure physically adapts before the user experiences a disruption.